

Applied Machine Learning

Feature Selection: Introduction



Learning goals

- Reasons for Feature Selection
- Types of Feature Selection



Recap

WHAT IS FEATURE ENGINEERING?

Feature engineering is on the intersection of **data cleaning**, **feature creation** and **feature selection**.

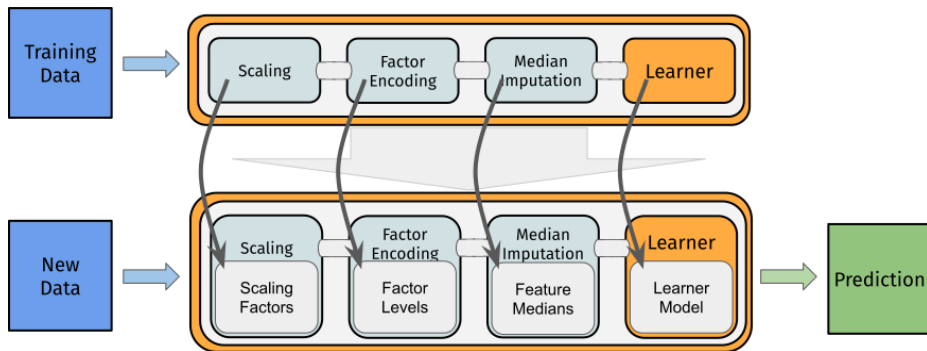
The goal is to solve common difficulties in data science projects, like

- skewed/weird feature distributions,
- (high cardinality) categorical features,
- functional (temporal) features,
- missing observations,
- high dimensional data,
- ...

and improve model performance (or make the data readable by the model in the first place).



TRAIN-TEST LEAKAGE



TYPES OF FEATURE ENGINEERING (SO FAR)

- numerical encoding of categorical features
- feature transformations
- imputation



Feature Selection



MOTIVATION



- Naive view:
 - More features $\rightarrow$ more information $\rightarrow$ discriminant power $\uparrow$
 - Model is not harmed by irrelevant features since their parameters can simply be estimated as 0.
- In practice, irrelevant and redundant features can “confuse” learners (see **curse of dimensionality**) and worsen performance.
- Example: In linear regression, R^2 is monotonically increasing in p , but adding irrelevant features leads to overfitting (capturing noise).

SIZE OF DATASETS

Many new forms of technical measurements and connected data leads to availability of extremely high-dimensional data sets.

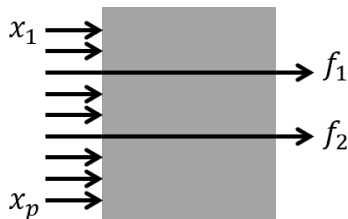
- **Classical setting:** Up to around 10^2 features, feature selection might be relevant, but benefits often negligible.
- **Datasets of medium to high dimensionality:** At around 10^2 to 10^3 features, classical approaches can still work well, while principled feature selection helps in many cases.
- **High-dimensional data:** 10^3 to 10^9 or more features. Examples: micro-array / gene expression data and text categorization (bag-of-words features). If we also have few observations, scenario is called $p \gg n$.



FEATURE SELECTION VS. EXTRACTION

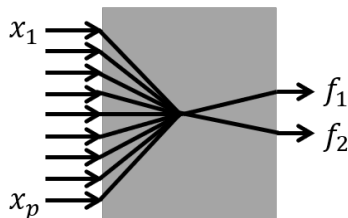


Feature selection



- Creates a subset of original features $\mathbf{x}$ by selecting $\tilde{p} < p$ features $\mathbf{f}$.
- Retains information on selected individual features.

Feature extraction



- Maps p features in $\mathbf{x}$ to $\tilde{p}$ extracted features $\mathbf{f}$.
- Info on individual features can be lost through (non-)linear combination.

REASONS FOR FEATURE SELECTION

- improved predictive performance, since we reduce overfitting on irrelevant features,
- robust models that do not rely on noisy features,
- simpler models that are easier to interpret,
- faster model fitting, e.g. for model updates,
- faster prediction, and
- no need to collect potentially expensive features.



GOALS OF FEATURE SELECTION



Single Feature Selection Problem: identifying a minimal-size subset of the variables that is optimally predictive for an outcome variable T of interest.

Multiple Feature Selection Problem: Let $\mathcal{S}$ be the solution to the single feature selection problem (called the reference solution). The solution $\mathcal{M}$ to the multiple feature selection problem consists of all minimal-size sets $\mathcal{S}_i \in \mathcal{F}$ that are statistically equivalent to $\mathcal{S}$.

- Important for knowledge discovery (know all possible solutions)
- We will not delve into this any further

TYPES OF FEATURE SELECTION METHODS



In rest of the chapter, we introduce different types of methods for FS:

- Filters: evaluate relevance of features using statistical properties such as correlation with target variable
- Wrappers: use a model to evaluate subsets of features
- Embedded methods: integrate FS directly into specific model - we look at them in their dedicated chapters (e.g., CART, L_0 , L_1)

Example: embedded method (Lasso) regularizing model params with L_1 penalty enables "automatic" feature selection:

$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \|\boldsymbol{\theta}\|_1 = \sum_{i=1}^n (y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x})^2 + \lambda \sum_{j=1}^p |\theta_j|$$